

Question 1:

Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development and video conferencing tasks.

For a startup team that performs **concurrent software development and video conferencing**, selecting the right **CPU, RAM, and storage** is essential to ensure **hardware compatibility** and avoid **performance bottlenecks**. The components should be chosen so that they work well together and meet both current and future workload requirements.

1. CPU (Central Processing Unit) Selection

The CPU is the main processing unit of the computer and greatly affects multitasking performance.

Selection Criteria:

- Choose a modern processor such as **Intel Core i5/i7 (12th generation or newer)** or **AMD Ryzen 5/7 (5000 series or newer)**.
- Prefer **6–8 cores and 12–16 threads** for smooth multitasking.
- Ensure the CPU supports virtualization if developers use virtual machines or Docker.
- Verify that the CPU is compatible with the motherboard socket (e.g., LGA1700 for Intel or AM5 for newer AMD processors).

Why it matters:

- Developers often run IDEs, browsers, databases, and containers simultaneously.
- Video conferencing applications (Zoom, Microsoft Teams, Google Meet) also require CPU resources.
- A multi-core processor prevents slowdowns when multiple applications run at the same time.

2. RAM (Memory) Selection

RAM temporarily stores data that active applications need.

Selection Criteria:

- Minimum **16 GB DDR4 or DDR5 RAM** per system.
- **32 GB RAM** is recommended for developers using virtual machines, Android Studio, or multiple development tools.
- Ensure the motherboard supports the RAM type (DDR4 or DDR5), speed, and maximum capacity.
- Use dual-channel memory (e.g., 2×8 GB or 2×16 GB) for improved performance.

Why it matters:

- Insufficient RAM causes frequent disk swapping, making the system slow.
- Adequate RAM allows developers to keep multiple applications open while attending video meetings without lag.

3. Storage Selection

Storage affects boot time, application loading, and file access speed.

Selection Criteria:

- Use an **NVMe SSD** instead of a traditional HDD.
- Minimum **512 GB SSD** for operating system, development tools, and projects.
- **1 TB SSD** is recommended for larger code repositories, virtual machines, and multimedia files.
- Ensure the motherboard has the required **M.2 NVMe slot** and supports the SSD generation (PCIe Gen3 or Gen4).

Why it matters:

- SSDs provide much faster boot and application loading times than HDDs.
- Faster storage improves compilation speed, file transfers, and overall responsiveness.
- Prevents delays when opening large projects or switching between applications.

Ensuring Hardware Compatibility

To ensure all components work together:

- Verify **CPU and motherboard socket compatibility**.
- Match **RAM type (DDR4/DDR5)** with motherboard specifications.
- Confirm motherboard supports the chosen **SSD interface (M.2 NVMe)**.
- Use a **power supply (PSU)** with sufficient wattage for all components.
- Check BIOS support for newer processors if required

Preventing Performance Bottlenecks

A balanced system avoids one component limiting the performance of others.

Component	Recommended Specification	Benefit
CPU	Intel Core i5/i7 or AMD Ryzen 5/7 (6–8 cores)	Smooth multitasking and faster code compilation
RAM	16 GB minimum, 32 GB recommended	Handles multiple applications without slowdown
Storage	512 GB–1 TB NVMe SSD	Fast boot, quick application loading, and rapid file access

